



Clesrovimab Sell-Side Analyst Call

October 22, 2024



Forward-looking statement of Merck & Co., Inc., Rahway, N.J., USA

This presentation of Merck & Co., Inc., Rahway, N.J., USA (the “company”) includes “forward-looking statements” within the meaning of the safe harbor provisions of the U.S. Private Securities Litigation Reform Act of 1995. These statements are based upon the current beliefs and expectations of the company’s management and are subject to significant risks and uncertainties. There can be no guarantees with respect to pipeline candidates that the candidates will receive the necessary regulatory approvals or that they will prove to be commercially successful. If underlying assumptions prove inaccurate or risks or uncertainties materialize, actual results may differ materially from those set forth in the forward-looking statements.

Risks and uncertainties include but are not limited to, general industry conditions and competition; general economic factors, including interest rate and currency exchange rate fluctuations; the impact of pharmaceutical industry regulation and health care legislation in the United States and internationally; global trends toward health care cost containment; technological advances, new products and patents attained by competitors; challenges inherent in new product development, including obtaining regulatory approval; the company’s ability to accurately predict future market conditions; manufacturing difficulties or delays; financial instability of international economies and sovereign risk; dependence on the effectiveness of the company’s patents and other protections for innovative products; and the exposure to litigation, including patent litigation, and/or regulatory actions.

The company undertakes no obligation to publicly update any forward-looking statement, whether as a result of new information, future events or otherwise. Additional factors that could cause results to differ materially from those described in the forward-looking statements can be found in the company’s Annual Report on Form 10-K for the year ended December 31, 2023 and the company’s other filings with the Securities and Exchange Commission (SEC) available at the SEC’s Internet site (www.sec.gov).

Agenda



Dr. Paula Annunziato

SVP, Head of Infectious Diseases and Vaccines, Global Clinical Development



Dr. Macaya Douoguih

VP, Vaccines, Global Clinical Development



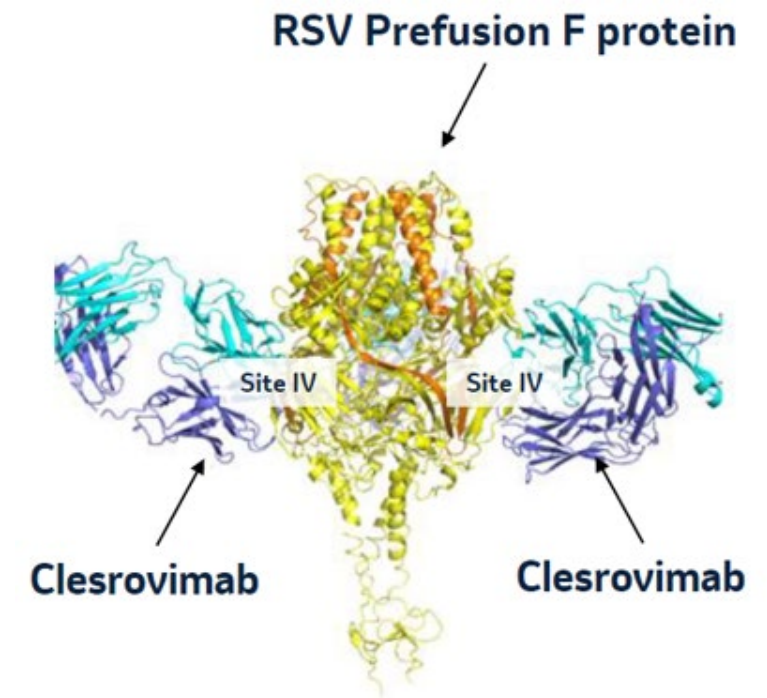
Art Hirt

SVP, Head of U.S. Vaccines

- > **Clesrovimab differentiation** | Dr. Paula Annunziato
- > **Protocol 004: Phase 2b/3 trial results** | Dr. Macaya Douoguih
- > **RSV patient dynamics and unmet need** | Arthur Hirt
- > **Q&A** | All

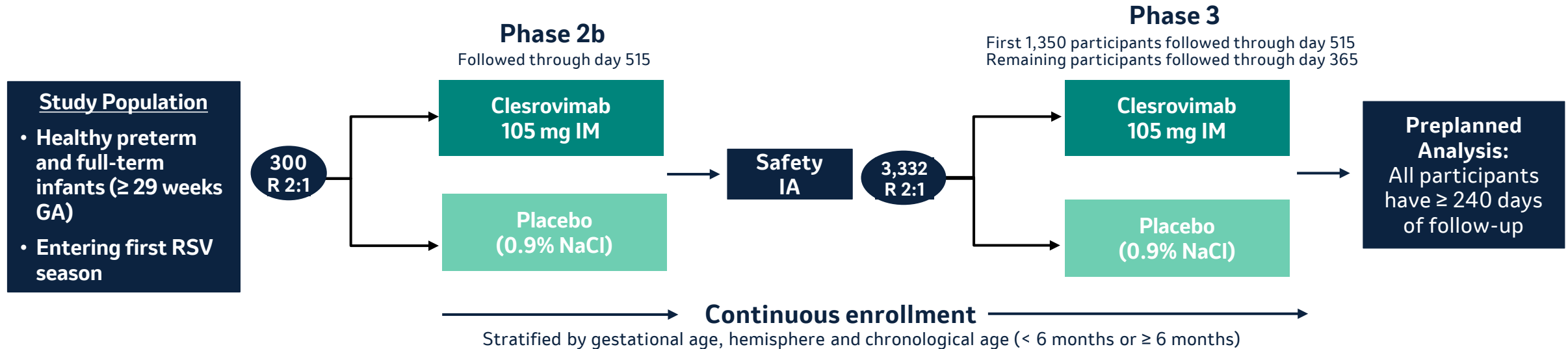
Clesrovimab is a differentiated candidate to help protect infants from RSV

- ✓ **Extended half-life mAb developed as a passive immunization for the prevention of RSV**
- ✓ **Same single dose injection for infants regardless of weight**
- ✓ **Protection through an infant's first full RSV season (5-6 months)**
- ✓ **Site of action (Site IV) has a low risk for development of resistance**



Protocol 004: Phase 2b/3 trial design

Randomized, Double-Blind, Placebo-Controlled, Pivotal Trial



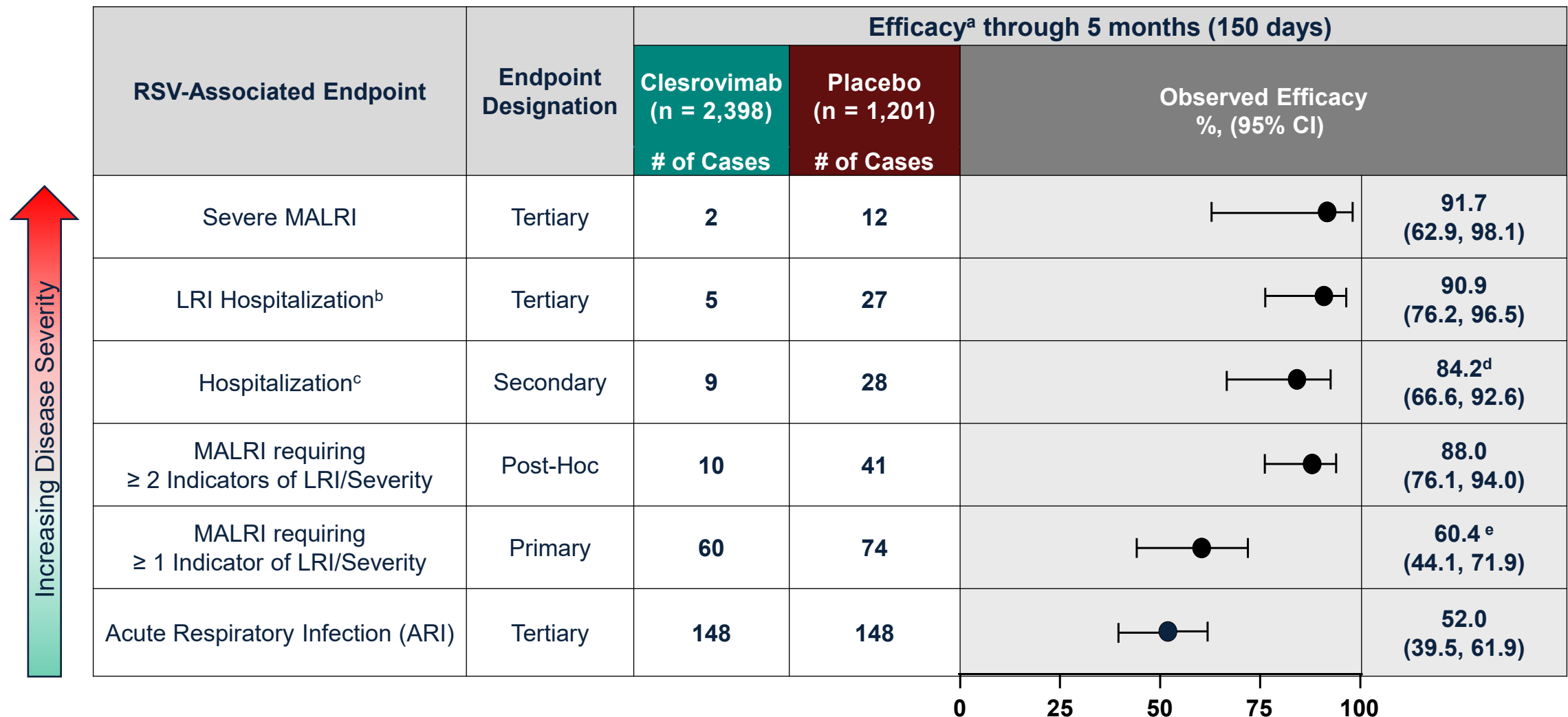
Primary endpoints:

- Safety and tolerability of clesrovimab compared to placebo
- Efficacy of clesrovimab against RSV-associated MALRI requiring ≥ 1 indicators of LRI or severity days 1-150
 - Hypothesis tested: Statistical criterion for success requires the LB of the 95% CI to be >25%

Secondary endpoints:

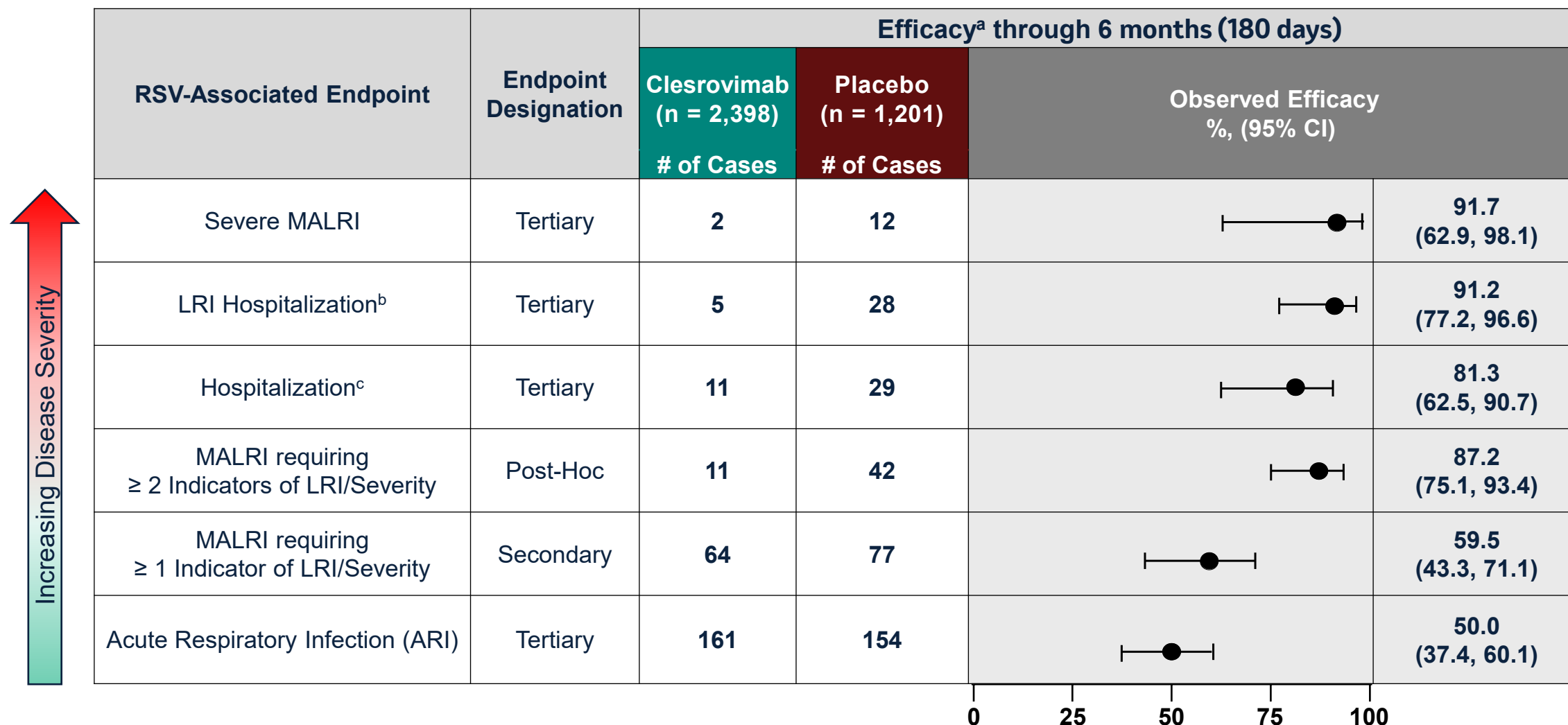
- Efficacy of clesrovimab against RSV-associated hospitalization, days 1-150
 - Hypothesis tested: Statistical criterion for success requires the LB of the 95% CI to be >0%
- Efficacy of clesrovimab against RSV-associated MALRI requiring ≥ 1 indicators of LRI or severity days 1-180

Protocol 004: a single dose of clesrovimab protected against RSV disease through 5 months



a. Full Analysis Set Population. b. LRI Hospitalization defined as: RSV PCR positive AND hospital admission for respiratory illness AND cough or difficulty breathing AND at least 1 of the following: wheezing, chest wall indrawing/retractions, rales/crackles, hypoxemia, tachypnea, dehydration due to respiratory symptoms. c. RSV-associated Hospitalization defined as: RSV PCR positive AND hospital admission for respiratory illness. d. Secondary endpoint, $p < 0.001$ (criterion=lower bound of the 95% CI >0%). e. Primary endpoint, $p < 0.001$ (criterion=lower bound of the 95% CI >25%).

Protocol 004: met all prespecified endpoints with data through 6-months



a. Full Analysis Set Population. b. LRI Hospitalization defined as: RSV PCR positive AND hospital admission for respiratory illness AND cough or difficulty breathing AND at least 1 of the following: wheezing, chest wall indrawing/retractions, rales/crackles, hypoxemia, tachypnea, dehydration due to respiratory symptoms. c. RSV-associated Hospitalization defined as: RSV PCR positive AND hospital admission for respiratory illness. d. Secondary endpoint, p<0.001 (criterion=lower bound of the 95% CI >0%). e. Primary endpoint, p<0.001 (criterion=lower bound of the 95% CI >25%).

Clesrovimab had a safety profile generally comparable to placebo

	Clesrovimab N=2409	Placebo N=1202	% Difference vs. placebo
	n (%)	n (%)	Estimate, (95% CI)
Overall AEs (Days 1-365)			
≥1 AE	1816 (75.4)	918 (76.4)	-1.0 (3.9, 2.0)
Drug-related AEs ^a	587 (24.4)	296 (24.6)	-0.3 (-3.3, 2.7)
SAE	278 (11.5)	149 (12.4)	-0.9 (-3.2, 1.3)
Drug-related SAE ^a	1 (0.0)	1 (0.1)	-0.0 (-0.4, 0.2)
Deaths	7 (0.3)	3 (0.2)	0.0 (-0.5, 0.4)
Solicited AEs and Fever (Days 1-5)			
Injection site pain	122 (5.1)	77 (6.4)	-1.3 (-3.1, 0.2)
Injection site erythema	90 (3.7)	40 (3.3)	0.4 (-0.9, 1.6)
Injection site swelling	65 (2.7)	31 (2.6)	0.1 (-1.1, 1.2)
Irritability	450 (18.7)	237 (19.7)	-1.0 (-3.8, 1.7)
Somnolence	303 (12.6)	171 (14.2)	-1.6, (-4.1, 0.7)
Decreased appetite	106 (4.4)	61 (5.1)	-0.7 (-2.3, 0.7)
Fever ^b	13 (0.5)	14 (1.2)	-0.6 (-1.4, 0.7)
AESI (Days 1-42)			
Anaphylaxis/hypersensitivity AESI	1 (0.0)	0 (0.0)	0.0 (-0.3, 0.2)
Rash AESI	11 (0.5)	4 (0.3)	0.1 (-0.4, 0.5)

N=All Participants As Treated Population. Estimated differences and CIs were calculated based on the Miettinen & Nurminen method.
 1) nonserious adverse events that occurred from Days 1 through 42 and 2) serious adverse events that occurred from visits Days 1 through 365 postdose and also include any reported serious adverse events after completion of participation in the study.
^aRelatedness to treatment was determined by investigator. ^bFever defined as rectal temperature ≥102.2°F or axillary temperature ≥101.7°F.

Key Takeaways

- The **majority of AEs were grade 1 or 2** (mild or moderate)
- **All AESI events were non-serious** and Grade 1 or 2, except for one Grade 3 event of urticaria in the clesrovimab group not considered related to study intervention by the investigator
- No deaths were considered related to study intervention by the investigator or attributed to RSV

Clesrovimab program conclusions



Efficacy

Clesrovimab, administered as a **single dose, regardless of weight, helped protect against a range of RSV disease** for infants entering their first RSV season, including term, preterm, and those with risk factors

- Demonstrated a **significant reduction in incidence of RSV disease and hospitalization**, with no shifting of RSV MALRI disease burden in second RSV season (Protocol 004) in healthy infants
- Reduced RSV-associated hospitalizations by more than 84% (secondary endpoint) through 5 months
- Observed a reduction in RSV-associated lower respiratory infection hospitalizations **by more than 90%, (tertiary endpoint) through 6 months**
- **Sustained efficacy through the 6 month RSV season in preterm and healthy infants** (Protocol 004), providing durable efficacy for an entire typical RSV season



Safety

Clesrovimab had a **safety profile** generally comparable to placebo and was **consistent across infant populations in healthy infants and those at increased risk for RSV disease**

- Safety profile generally comparable to placebo **in healthy preterm and full-term infants** born during or entering their first RSV season
- Safety profile generally comparable to palivizumab in infants at **increased risk for severe RSV disease** and was consistent with the safety profile in healthy infants (Protocol 007, interim analysis)

RSV is the leading cause of hospitalization among infants

Significant unmet need



Global^{1,2}

RSV infections in children ≤ 1 year old cause
~**12.9 million lower respiratory infections** with:

- ~**2.2 million hospitalizations** per year,
most of which take place during the first 6
months of life



United
States³

RSV infections in children ≤ 1 year old cause :

- ~**1.7 million outpatient visits** per year
- ~**96,000 hospitalizations** per year

clesrovimab

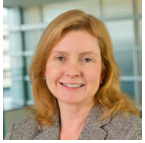
designed to protect infants for the
full 6-month RSV season

designed to provide the convenience of a
single dose, regardless of weight

represents a
blockbuster commercial opportunity



Q&A



Dr. Paula Annunziato
SVP, Head of Infectious Diseases and Vaccines,
Global Clinical Development



Dr. Macaya Douoguih
VP, Vaccines, Global Clinical Development



Art Hirt
SVP, U.S. Vaccines



Peter Dannenbaum
SVP, Investor Relations



Acronyms

AE = Adverse event

AESI = Adverse event of special interest

ARI = Acute respiratory infection

CI = Confidence interval

GA = Gestational age

IA = Interim analysis

IM = Intramuscular

LB = Lower bound

LRI = Lower respiratory tract infection

mAb = Monoclonal antibody

MALRI = Medically Attended Lower Respiratory Infection

R = Randomized

RSV = Respiratory syncytial virus

SAE = Serious adverse event

Paula Annunziato, MD

SVP, Head of Infectious Diseases and Vaccines, Global Clinical Development

Paula Annunziato, MD joined Merck Research Laboratories (MRL) in 2002 as Director, Biologics and Vaccine Clinical Research and has held positions of Product Development Team Leader, Section Head, and Therapeutic Area Head for Merck Vaccines. Most recently she was promoted to SVP ID and Vaccine Global Clinical Development.

Prior to joining Merck, Paula was Associate Professor of Clinical Pediatrics at Columbia University College of Physicians and Surgeons and a member of the Division of Pediatric Infectious Diseases. During her time at Columbia University, Paula's research focused on latency and reactivation of the varicella zoster virus (VZV).

Paula received her Medical Degree from the University of Texas Health Science Center in San Antonio, Texas, completed her Pediatric residency at the Schneider Children's Hospital/Long Island Jewish Medical Center in New Hyde Park, New York, and her Pediatric Infectious Diseases fellowship at the University of Rochester in Rochester, New York.



Macaya Douoguih, MD, MPH

VP, Vaccines, Global Clinical Development

Macaya Douoguih, MD, MPH joined Merck Research Laboratories (MRL) in December 2023 as Vice President and Therapeutic Area Head for Vaccine Global Clinical Development.

Prior to joining Merck, Macaya was Global Head of Late Development for the Infectious Diseases and Vaccines Therapeutic Area at Janssen. During her 13-year tenure, she also held positions of Head of Clinical Development, Clinical Franchise Leader for Emerging Diseases and Bio-preparedness, and Medical Leader largely focused on viral vectored and bacterial vaccine programs targeting tuberculosis, malaria, Ebola, HIV, SARS-CoV-2, E.coli and RSV, and monoclonal antibodies targeting influenza. Prior to joining Janssen, she has worked at Aeras, the NIH and PAHO.

Macaya received her Medical Degree from the University of Wisconsin-Madison, in Madison, Wisconsin, completed her Family Medicine residency at University of Maryland Medical System in Baltimore, Maryland, and her Preventive Medicine Residency and MPH at Johns Hopkins Bloomberg School of Public Health in Baltimore, Maryland.



Art Hirt

SVP, U.S. Vaccines

Art joined Merck in 1997 and over that time has held business leadership roles in the US market, global, and ex-US subsidiaries and regions. Prior to his current position, Art was the SVP, Hospital & Specialty Business Unit, after spending time in the US Oncology team with responsibility for leading the commercial customer-facing teams. His oncology experience also includes leading the Oncology Business Unit in the Latin America Region.

Other prior notable roles include VP/Chief Marketing Officer for MSD Japan in Tokyo, Sales VP in the US market, VP/Global Brand Leader for Gardasil, and VP of Marketing for the Merck/Schering-Plough joint venture. Prior to joining Merck, Art held various sales and marketing roles at Pfizer in New York.

Art has a Bachelor's degree from SUNY Plattsburg and a Masters of Marketing Research from the University of Georgia.

